

renAIssance: Printed OCR

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Project: RenAIssance: Improving OCR for Early Modern Printed Spanish Sources

Synopsis

This project aims to build an end-to-end OCR pipeline designed specifically for 17th-century printed Spanish sources. Traditional OCR engines and modern vision-language models struggle with historical typefaces, such as the long-s, archaic ligatures, and abbreviation marks. This proposal details a custom CNN-RNN model trained with inverse-frequency weighted CTC loss to handle rare diacritics, paired with a lexicon-constrained beam search decoder and Gemini LLM post-correction. The goal is to reduce the Character Error Rate to a usable level for digital humanities researchers on limited labelled data.

Benefits to Community

There are thousands of 17th-century Spanish documents in archives that are essentially inaccessible to computational analysis because the OCR quality is too low to be useful. A pipeline that can get CER below 0.3 on clean pages, which I think is achievable with proper fine-tuning, would make those documents searchable, linkable, and analyzable in ways that are not currently possible. This project will deliver a fully open-source pipeline that Google, HumanAI, and society at large can use to unlock historical Spanish texts.

Deliverables

The main gap right now is that CRNN training does not work properly at the page level. Fixing this requires per-line crops with per-line text labels, which means better line segmentation.

Phase 1: Better data and line-level training (Weeks 1 to 4) * Train a lightweight trainable line detector (U-Net or YOLO) fine-tuned on annotated rows. **[Required]** * Expand ground truth coverage by transcribing the remaining 19 unassigned pages. **[Required]**

Phase 2: Model improvements (Weeks 5 to 8) * End-to-end line-level CRNN training with the new data. **[Required]** * Experiment with self-supervised pre-training on unlabelled page crops. **[Optional]** * Replace BiLSTM head with a lightweight cross-attention layer. **[Optional]**

Phase 3: System integration and evaluation (Weeks 9 to 12) * Full pipeline evaluation across all sources. **[Required]** * Clean public-facing demo and documentation. **[Required]**

Timeline

Week	Work	Status
1	Set up communication with mentors, finalize ground truth expansion plan, annotate 5 to 10 pages for line detection	Required
2 to 3	Train lightweight line detector, replace projection heuristic with model-based segmentation	Required
4	End-to-end CRNN training at line level, verify loss curves look healthy	Required
5 to 6	Experiment with augmentation, try cross-attention CRNN variant	Optional
7	Midterm evaluation, document results so far	Required
8	Self-supervised pre-training experiments	Optional
9 to 10	Full pipeline evaluation across all 6 sources	Required
11	Fine-tuned TrOCR vs CRNN comparison, statistical analysis	Required
12	Final documentation, demo, cleanup, final evaluation	Required

Related Work

Existing OCR engines like Tesseract and Kraken are widely used in digital humanities but often require extensive fine-tuning and layout annotation to work well on historical texts. Modern vision-transformer approaches like TrOCR provide a strong zero-shot baseline but still suffer from domain mismatch on 17th-century letterforms. This project builds upon these foundations by combining a lightweight, domain-specific CRNN architecture with modern LLM post-processing (Gemini), demonstrating that a hybrid approach—specialized recognition and contextual correction—can significantly outperform zero-shot monolithic models.

For the evaluation test, I put together a full preliminary pipeline before submitting this proposal. Results on 3 matched evaluation pages showed Gemini LLM cleanup reduced CER from 1.1468 (TrOCR zero-shot) to 1.1026. The pipeline code is at: <https://github.com/Magma4/renaissance-ocr>

Biographical Information

I am a Computer Science student with a strong interest in machine learning applied to cultural heritage problems. I have been working with computer vision

and NLP for a couple of years, not just as coursework but because I find the problem of making historical documents machine-readable genuinely interesting. Most OCR research focuses on modern text, so the failure modes on 17th-century typefaces are actually a different and underexplored problem.

I am comfortable working in Python, PyTorch, and Transformers. I have built sequence-to-sequence models before and I know the CTC loss formulation well enough to have debugged it from scratch. I can work independently and I know when to ask questions.

Practical Notes

This runs as a standard 175-hour GSoC project. I have a MacBook with an M-series chip for development. Compute-heavy training I will run on a cloud VM where I have GCP credits. I am available about 15 hours per week during the 12-week GSoC period to complete the 175 hours. I prefer async communication but am happy to do weekly video syncs if that works for the mentors.